

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – 4.2 PROBABILITY I: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How is probability applied in real life situations? You have worked with the KMD forecast of a 70% chance of rain in Kericho and the Rift Valley on a March morning before the long rains. Wanjiru Chebet (a maize farmer near Eldoret with 2 acres), Captain Otieno (a pilot at JKIA flying to Kisumu) and Ms Achieng (Nairobi County disaster office) must each decide what to do. Write a Final Explanation in your own words using Claim, Evidence and Reasoning. Use the numbers and methods from Lessons 1 to 9. Show your working for every calculation and write probabilities as fractions or decimals between 0 and 1. Answer each section in the space under its prompt.

Section 1: What does a 70% forecast mean?

The KMD forecasts a 70% chance of rain. (a) Does this guarantee rain on that day? Give a claim. (b) Use the Kericho March record (rain on 14 days out of 20) to show how a number like 0.7 can be worked out. (c) In the coin activity, the running relative frequency of heads was 0.6 after 10 tosses, 0.55 after 20, 0.55 after 40 and 0.525 after 80 tosses. Explain what this pattern tells you about trusting estimates from long records.

(a) Claim: No. A 70% chance of rain does not guarantee rain. It means that in a long run of similar days, about 7 out of every 10 have rain, so a single day can still be dry. (b) Experimental probability (relative frequency) = number of times the event occurred / total number of trials. For Kericho, rain occurred on 14 of 20 days, so $14/20 = 0.7$, which is 70%. (c) The relative frequency of heads changed from 0.6 to 0.55, 0.55 and 0.525 as the number of tosses grew from 10 to 80, and the changes became smaller, so the value became steadier and closer to the fair-coin value of 0.5. More trials give a more stable, more trustworthy estimate. So a forecast based on a long record, like Kericho's, can be trusted as a long-run likelihood even though one day may surprise us.

Section 2: Sample space, events and theoretical probability

(a) List the sample space S for the weather on one March day in Kericho, using the three day states from class. (b) Write the event 'rain, as the farmer sees it' as a subset of S. (c) For a fair

(a) $S = \{\text{rain, storm, dry}\}$. (b) 'Rain, as the farmer sees it' = $\{\text{rain, storm}\}$. (c) Theoretical probability = favourable outcomes / total equally likely outcomes. For a fair coin $S = \{H, T\}$, so $P(H) = 1/2 = 0.5$. For a fair die $S = \{1, 2, 3, 4, 5, 6\}$ and even = $\{2, 4, 6\}$, so $P(\text{even}) = 3/6 = 1/2$. (d) The experimental result $42/80 = 0.525$ is close to 0.5 but not equal because of chance variation in a limited number of trials. The difference tends to shrink as the number of trials grows. (e) No. For the die each outcome is equally likely, but rain, storm and dry are not equally likely, so counting 1 out of 3 would be wrong. Their probabilities, $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$, must come from records and models. They add up to $0.6 + 0.1 + 0.3 = 1.0$, which is the check that they cover

coin and for a fair die, calculate $P(H)$ and $P(\text{even number})$ using theoretical probability. (d) Our class got 42 heads in 80 tosses (0.525) but theory gives 0.5. Explain why they differ. (e) Can we find $P(\text{rain})$ for the weather by counting outcomes in S in the same way as for the die? Explain.	the whole sample space.
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Section 3: Combining probabilities on one day

(a) Is 'rain' and 'dry' on the same day at the same place mutually exclusive? Give a reason. (b) Use $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$ to find $P(\text{rain only or storm})$ and $P(\text{storm or dry})$. Name the law you used and say why it is allowed. (c) Find $P(\text{no rain})$ using the complement rule. (d) How does your answer to (b) agree with the KMD forecast of 70%?	(a) Yes. Rain and dry cannot both happen on the same day at the same place, so no outcome is in both events and $P(\text{rain and dry}) = 0$. (b) Addition law for mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B)$. $P(\text{rain only or storm}) = 0.6 + 0.1 = 0.7$. $P(\text{storm or dry}) = 0.1 + 0.3 = 0.4$. The law is allowed because the day states cannot happen together. (c) Complement rule: $P(\text{not } A) = 1 - P(A)$. $P(\text{no rain}) = 1 - 0.7 = 0.3$. (d) The forecast of 70% is the chance of rain as the farmer sees it, which is rain only (0.6) plus storm (0.1) = 0.7. The remaining 0.3 is the chance of no rain.
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Section 4: Two days of weather

Wanjiru wants to know about the next two days. Assume the days are independent, with $P(R) = 0.7$ and $P(N) = 0.3$ on each day. (a) Why is the multiplication law suitable here? (b) Draw a probability tree diagram for two days and write the probability of each of the four outcomes. Check that they add to 1. (c) Find $P(\text{exactly one rainy day})$ and $P(\text{at least one rainy day})$. Show a second way of getting $P(\text{at least one rainy day})$. (d) Why is $P(\text{rain on both days})$ less than 0.7?	(a) Rain on one day does not change the chance of rain on the next day, so the days are independent and $P(A \text{ and } B) = P(A) \times P(B)$ can be used. (b) Tree: Day 1 branches R (0.7) and N (0.3); from each, Day 2 branches R (0.7) and N (0.3). Multiply along branches: $P(RR) = 0.7 \times 0.7 = 0.49$; $P(RN) = 0.7 \times 0.3 = 0.21$; $P(NR) = 0.3 \times 0.7 = 0.21$; $P(NN) = 0.3 \times 0.3 = 0.09$. Total = $0.49 + 0.21 + 0.21 + 0.09 = 1.00$. (c) Exactly one rainy day = $RN \text{ or } NR = 0.21 + 0.21 = 0.42$. At least one rainy day = $0.49 + 0.21 + 0.21 = 0.91$. Second way: complement, $1 - P(NN) = 1 - 0.09 = 0.91$. (d) $P(RR) = 0.49$ is less than 0.7 because both events must happen, and multiplying two numbers less than 1 gives a smaller number. About 49 out of every 100 such pairs of days would have rain on both days.
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both days) smaller than $P(\text{rain on one day})$?	
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Section 5: Final Explanation – advising the decision makers	
Using the class decision rules, advise each person. Farmer Wanjiru plants if $P(\text{rain on each of the next two days}) \geq 0.49$ is considered good, or if $P(\text{at least one rainy day in two}) \geq 0.9$. Captain Otieno delays or reroutes if $P(\text{storm}) \geq 0.1$. Ms Achieng prepares drainage and warns residents if $P(\text{rain}) \geq 0.7$ or $P(\text{storm}) \geq 0.1$. (a) State what each person should do and give the probability evidence. (b) Write a Claim, Evidence and Reasoning answer to the driving question: How is probability applied in real life situations? In your reasoning explain how the numbers are worked out and combined, and why people can still act sensibly if the day turns out differently.	(a) Wanjiru: $P(RR) = 0.49$ meets the 0.49 rule and $P(\text{at least one rainy day}) = 0.91$ is above 0.9, so she should prepare to plant and keep irrigation as a backup. Captain Otieno: $P(\text{storm}) = 0.1$ meets the threshold of 0.1, so he should plan for a possible storm, and check again before take-off, delaying or rerouting if needed. Ms Achieng: $P(\text{rain}) = 0.7$ meets the 0.7 rule and $P(\text{storm}) = 0.1$ meets the 0.1 rule, so the office should prepare drainage and warn residents. These thresholds are classroom decision rules, not real KMD rules. (b) Claim: Probability is applied by turning past records and models into numbers between 0 and 1 that people compare with the cost of being wrong. Evidence: Kericho had rain on 14 of 20 March days, so relative frequency = $14/20 = 0.7$, matching the forecast. The day states have $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$, which sum to 1.0, and the addition law gives $0.6 + 0.1 = 0.7$. The multiplication law and tree give $P(RR) = 0.49$ and $P(\text{at least one rainy day}) = 0.91$. Reasoning: A forecast of 70% does not guarantee rain. It describes the long run, about 7 out of 10 similar days, and long records give steadier estimates. Listing the sample space, classifying events as mutually exclusive or independent, and then using the addition law, the multiplication law, the complement rule and tree diagrams lets us combine the numbers. A decision based on probability can be unlucky on one day, for example the 0.09 chance of two dry days, but it is the best choice over many days.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Meaning of probability and relative frequency	Clearly explains that 70% is a long-run likelihood, not a guarantee, and uses $14/20 = 0.7$ and the running relative frequencies correctly to show that more trials give steadier estimates.	States that the forecast is not a guarantee and calculates $14/20 = 0.7$ correctly, but gives limited explanation of why more trials matter.	Treats 70% as a guarantee or does not link it to past records or relative frequency.
Sample space, events and theoretical probability	Lists $S = \{\text{rain, storm, dry}\}$, writes events as subsets, calculates $P(H)$ and $P(\text{even})$ correctly, and explains both chance variation and why the weather states are not equally likely.	Lists the sample space and calculates theoretical probabilities correctly, with a partial explanation of differences from experimental results.	Sample space is incomplete or theoretical probabilities are incorrect, with no explanation of differences.

Use of the laws of probability	Chooses the addition law, complement rule and multiplication law correctly, justifies each by mutually exclusive or independent events, and calculates 0.7, 0.4, 0.3 and 0.49 correctly.	Applies the laws correctly with minor gaps in justifying why each one is allowed.	Mixes up the laws or applies them without checking the events, giving incorrect answers.
Tree diagram and two-day probabilities	Draws a fully labelled tree, multiplies along branches, checks that the total is 1.00, and finds 0.42 and 0.91 with the complement as a check.	Draws a correct tree and finds most of the outcomes correctly, with a missing check or small error.	Tree is incomplete or probabilities are not multiplied and added correctly.
Claim, Evidence, Reasoning and decisions	Gives a clear claim, uses specific evidence from the lessons, applies thresholds correctly for Wanjiru, Captain Otieno and Ms Achieng, and explains why decisions based on probability are sensible despite occasional bad luck.	Gives a claim with some evidence and mostly correct decisions, but the reasoning is brief.	Claim is unclear, evidence is missing or decisions are not linked to the probabilities.

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